Code-Layout, Readability and Reusability

Date	08 July 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform indentation throughout the project	Yes	Standard Python 4-space indentation used throughout the project.
2	Proper File Structure	Files are logically organized	Yes	Project is organized into modules such as main.py, qna.py, summary_module.py, quiz_module.py, learning_path.py, templates and static folders.
3	Meaningful Variable Names	Variables clearly represent their purpose	Yes	Variables such as question, topic, summary, and learning_path improve readability.
4	Function / Method Names	Functions are descriptive	Yes	Functions like answer_question(), generate_quiz(), summarize_text(), and generate_learning_path() clearly indicate their purpose.
5	Code Comments	Comments explain important logic	Yes	Important sections are documented with comments for easier understanding and maintenance.
6	Modular Design	Code divided into reusable modules	Yes	Each AI feature is implemented in a separate Python module, making the project modular and reusable.

7	No Redundant Code	Duplicate code minimized	Yes	Reusable functions reduce repetition and improve maintainability.
8	Error Handling	Exceptions handled properly	Yes	Try-except blocks handle API failures and invalid user inputs gracefully.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Question Answer Module	Python, Google Gemini API	Answers student questions from any subject	High
2	Explanation Module	Python, Google Gemini API	Generates detailed explanations for concepts	High
3	Summarization Module	Python, Google Gemini API	Summarizes long paragraphs and notes	High
4	Quiz Generation Module	Python, Google Gemini API	Generates quizzes for different subjects	High
5	Learning Path Module	Python, Google Gemini API	Creates personalized learning roadmaps	High

Overall Code Quality Assessment:

S.No	Component / Module Name	Language / Technology
1	Question Answer Module	Python, Google Gemini API
2	Explanation Module	Python, Google Gemini API
3	Summarization Module	Python, Google Gemini API
4	Quiz Generation Module	Python, Google Gemini API
5	Learning Path Module	Python, Google Gemini API